

Gig Worker Earnings Reality Check

Complete step-by-step instructions to build the full dashboard in Microsoft Power BI Desktop — from CSV import to published report

What's included	Detail
CSV files	6 tables — pre-cleaned, pre-computed, ready to import
Relationships	Exact join keys and cardinality for every table pair
DAX measures	20 measures covering all 4 dashboard pages
Visual guide	Page-by-page: which visual, which fields, which format
Interview DAX	10 DAX questions with answers for interviews
Time to build	~45 minutes in Power BI Desktop (Free)

4 Dashboard Pages

Page	Title	Key Visuals
1	Earnings Overview	Waterfall · Line trend · Rain seasonality bar
2	City & Platform	Clustered bars · Matrix heatmap · Donut
3	Real Wage Erosion	Multi-line chart · 100% bar · KPI cards
4	Break-Even Calculator	Slicer-driven KPI cards · Bar · Gauge

SECTION 1 — The 6 CSV Files

You have 6 CSV files. Import all 6 into Power BI Desktop. Here is what each contains and why it exists.

File	Rows × Cols	Key columns added	Use in dashboard
Worker_Profiles.csv	300 × 14	engine_type, engine_value	Slicer filters, profile joins
Monthly_Earnings.csv	18,000 × 29	real_net_inr, fixed_costs_inr, breakeven_days, take_home_pct, is_monsoon, rain_bucket, year_month	Main fact table — all measures
City_Year_Summary.csv	15 × 10	real_wage_change_pct	Page 3 real wage line chart
Platform_Commission_Rates.csv	25 × 5	—	Commission trend line chart
Petrol_Prices_Monthly.csv	56 × 7	year_month	Petrol price context
Date_Table.csv	60 × 7	sort_order, quarter	Time axis for all charts

How to Import

Step 1 — Open Power BI Desktop (free download from microsoft.com/powerbi)

Step 2 — Home → Get Data → Text/CSV → select Worker_Profiles.csv → Load

Step 3 — Repeat for all 6 CSV files

Step 4 — After loading all 6, go to Transform Data (Power Query)

Step 5 — In Monthly_Earnings: set year and month columns to Whole Number data type

Step 6 — In Date_Table: set sort_order to Whole Number. Close & Apply.

Note: Monthly_Earnings is your FACT table. All other tables are DIMENSION tables.

SECTION 2 — Table Relationships

Go to Model View (the diagram icon on the left sidebar). Create these 5 relationships by dragging and dropping.

From Table	From Column	→	To Table	To Column	Cardinality	Filter Direction
Monthly_Earnings	worker_id	→	Worker_Profiles	worker_id	Many → One	Single
Monthly_Earnings	year_month	→	Date_Table	year_month	Many → One	Single
Monthly_Earnings	year_month	→	Petrol_Prices_Monthly	year_month	Many → One	Single
Platform_Commission_Rate	platform	→	Monthly_Earnings	platform	One → Many	Single
City_Year_Summary	year	→	Date_Table	year	Many → One	Single

Note: If Power BI auto-detects relationships, check they match this table exactly. Delete any wrong ones.

Tip: Set Date_Table as the Date Table: right-click it in Model View → Mark as Date Table → year_month column.

SECTION 3 — All DAX Measures (20 total)

Go to Report View. Click on Monthly_Earnings table in the Fields pane. Click New Measure. Copy-paste each formula below.

Note: Create a dedicated Measures table: Home → Enter Data → name it 'Measures' → Load. Store all measures there.

Group A — Core Earnings

// Avg Monthly Gross

```
Avg Monthly Gross = AVERAGE(Monthly_Earnings[gross_earnings_inr])
```

// Avg Monthly Net

```
Avg Monthly Net = AVERAGE(Monthly_Earnings[net_earnings_inr])
```

// Avg Take-Home Pct

```
Avg Take-Home Pct =  
DIVIDE(  
AVERAGE(Monthly_Earnings[net_earnings_inr]),  
AVERAGE(Monthly_Earnings[gross_earnings_inr])  
) * 100
```

// Total Net Earnings

```
Total Net Earnings =  
SUM(Monthly_Earnings[net_earnings_inr])
```

// Avg Hourly Rate

```
Avg Hourly Rate =  
AVERAGE(Monthly_Earnings[effective_hourly_rate_inr])
```

Group B — Deductions

// Avg Commission INR

```
Avg Commission INR = AVERAGE(Monthly_Earnings[commission_deducted_inr])
```

// Avg Commission Pct

```
Avg Commission Pct = AVERAGE(Monthly_Earnings[platform_commission_pct])
```

// Avg Fixed Costs

```
Avg Fixed Costs =  
AVERAGE(Monthly_Earnings[fuel_cost_inr])  
+ AVERAGE(Monthly_Earnings[phone_data_inr])  
+ AVERAGE(Monthly_Earnings[vehicle_maintenance_inr])
```

// Avg Rain Loss INR

```
Avg Rain Loss INR = AVERAGE(Monthly_Earnings[rain_income_loss_inr])
```

// Monsoon vs Dry Rain Ratio

```
Monsoon vs Dry Rain Ratio =  
VAR MonsoonLoss =  
CALCULATE(  
AVERAGE(Monthly_Earnings[rain_income_loss_inr]),  
Monthly_Earnings[is_monsoon] = "Monsoon"  
)
```

```
VAR DryLoss =  
CALCULATE(  
AVERAGE(Monthly_Earnings[rain_income_loss_inr]),  
Monthly_Earnings[is_monsoon] = "Non-Monsoon"  
)  
RETURN  
DIVIDE(MonsoonLoss, DryLoss)
```

Group C — Real Wage & Inflation

```
// Avg Real Net INR
Avg Real Net INR =
AVERAGE(Monthly_Earnings[real_net_inr])

// Real Wage Index (2021 = 100)
Real Wage Index =
VAR Base2021 =
CALCULATE(
AVERAGE(Monthly_Earnings[real_net_inr]),
Monthly_Earnings[year] = 2021
)
RETURN
DIVIDE(AVERAGE(Monthly_Earnings[real_net_inr]), Base2021) * 100

// Nominal Wage Index (2021 = 100)
Nominal Wage Index =
VAR Base2021 =
CALCULATE(
AVERAGE(Monthly_Earnings[net_earnings_inr]),
Monthly_Earnings[year] = 2021
)
RETURN
DIVIDE(AVERAGE(Monthly_Earnings[net_earnings_inr]), Base2021) * 100

// Real Wage Erosion Pct
Real Wage Erosion Pct =
VAR Current = AVERAGE(Monthly_Earnings[real_net_inr])
VAR Base2021 =
CALCULATE(
AVERAGE(Monthly_Earnings[real_net_inr]),
Monthly_Earnings[year] = 2021
)
RETURN
DIVIDE(Current - Base2021, Base2021) * 100
```

Group D — Break-Even

```
// Avg Breakeven Days
Avg Breakeven Days =
AVERAGE(Monthly_Earnings[breakeven_days])

// Avg Net Per Day
Avg Net Per Day =
AVERAGE(Monthly_Earnings[net_per_day])

// Monthly Profit After Fixed Costs
Monthly Profit After Fixed Costs =
AVERAGE(Monthly_Earnings[net_earnings_inr])
- AVERAGE(Monthly_Earnings[fixed_costs_inr])

// Profit Window Days
```

```
Profit Window Days =  
AVERAGE(Monthly_Earnings[working_days])  
- AVERAGE(Monthly_Earnings[breakeven_days])
```

Group E — Context KPIs

```
// Workers Count
```

```
Workers Count =  
DISTINCTCOUNT(Monthly_Earnings[worker_id])
```

```
// Best Platform Net
```

```
Best Platform Net =  
MAXX(  
SUMMARIZE(Monthly_Earnings, Monthly_Earnings[platform],  
"PlatformNet", AVERAGE(Monthly_Earnings[net_earnings_inr])),  
[PlatformNet]  
)
```

SECTION 4 — Build Each Dashboard Page

Page 1 — Earnings Overview

Right-click on the Pages panel → Add Page → rename to 'Earnings Overview'

Visual	Fields / Settings	Purpose
Waterfall chart (top left)	Category: drag a blank column or use a custom order table Breakdown: commission_deducted_inr, fuel_cost_inr, phone_data_inr, vehicle_maintenance_inr, rain_income_loss_inr Y-axis: gross_earnings_inr Use 'Avg Monthly Gross' as start measure	Shows where gross goes. Set Increase colour = green, Decrease = red, Total = blue
Line chart (top right)	X-axis: Date_Table[year] Line 1: Avg Monthly Gross Line 2: Avg Monthly Net Secondary Y: Avg Commission Pct	Gross vs Net 2021-2025 with commission overlay
Bar chart (bottom)	X-axis: Date_Table[month_name] Sort by: month number (use sort_order) Y-axis: Avg Rain Loss INR Conditional format: highlight months 6-9	Rain seasonality. Manually colour bars 6-9 blue
3 KPI cards (top strip)	Card 1: Avg Monthly Net Card 2: Avg Take-Home Pct Card 3: Avg Breakeven Days	Headline numbers at a glance
Slicer (left)	Field: Monthly_Earnings[city] Style: Dropdown or List	Filter whole page by city

Tip: For the waterfall, Power BI needs the breakdown fields individually — not the net field. Net is calculated automatically as the total minus all breakdowns.

Page 2 — City & Platform

Visual	Fields / Settings	Purpose
Clustered bar chart (top left)	Y-axis: city X-axis: Avg Monthly Net Legend: none Data labels: ON	City ranking by net earnings
Clustered bar chart (top right)	Y-axis: platform X-axis: Avg Monthly Gross + Avg Monthly Net (two series, same axis) Data labels: show Avg Commission Pct as tooltip	Platform gross vs net. Sort by Avg Monthly Net descending
Matrix visual (bottom left)	Rows: city Columns: platform Values: Avg Monthly Net Conditional formatting: Background colour scale Min=red Max=green	Platform x City heatmap. This is the crown jewel visual
Donut chart (bottom right)	Legend: vehicle_type (from Worker_Profiles) Values: Avg Monthly Net	Net by vehicle type
2 Slicers	Slicer 1: year Slicer 2: vehicle_type	Filter context for all visuals

Tip: For the matrix heatmap: go to Format → Conditional Formatting → Background Colour → Field Value → Avg Monthly Net. Set colour scale Red-Yellow-Green. This makes it look like a real heatmap.

Page 3 — Real Wage Erosion

Visual	Fields / Settings	Purpose
Line chart (main, large)	X-axis: Date_Table[year] Line 1: Real Wage Index (2021=100) Line 2: Nominal Wage Index (2021=100) Line 3: hard-code CPI as a What-If parameter or add City_Year_Summary[real_wage_index_2021_100]	The flagship chart. Shows real vs nominal divergence
Clustered bar chart	X-axis: year Bar 1: CPI rise (= City_Year_Summary[avg_commission_pct]) Bar 2: Real Wage Erosion Pct (two series side by side)	Year-by-year decomposition. CPI bar vs real wage bar
Line chart (commission)	X-axis: Platform_Commission_Rates[year] Line per platform: commission_rate_pct Legend: platform	Commission creep 2021-2025. All platforms rising
4 KPI Cards	Card 1: Real Wage Erosion Pct Card 2: Avg Commission Pct Card 3: Avg Real Net INR Card 4: Nominal Wage Index	Top strip summary numbers

Page 4 — Break-Even Calculator

Visual	Fields / Settings	Purpose
Slicer — City	Field: city (from Monthly_Earnings) Style: Tile (horizontal buttons)	Select city — drives all cards
Slicer — Vehicle	Field: vehicle_type (from Worker_Profiles) Style: Tile	Select vehicle type
Slicer — Year	Field: year Style: Dropdown or slider	Select year (2021–2025)
KPI Card — Break-Even	Measure: Avg Breakeven Days Format: 0.0 Label: 'Days to Break Even'	The main output number
KPI Card — Profit	Measure: Monthly Profit After Fixed Costs Format: Rs. #,##0 Label: 'Monthly Profit'	Profit after fixed costs
KPI Card — Window	Measure: Profit Window Days Label: 'Profit Window (Days)'	Days actually in profit
KPI Card — Fixed Costs	Measure: Avg Fixed Costs Label: 'Monthly Fixed Costs'	Total fuel+phone+maintenance
Clustered bar chart	Y-axis: vehicle_type X-axis: Avg Breakeven Days Small multiples: city Data labels: ON, 1 decimal	All scenarios at once. Sorted ascending (shorter = better)
Line chart	X-axis: year Lines: one per vehicle_type Values: Avg Breakeven Days Legend: vehicle_type	Break-even trend 2021-2025

Tip: The three slicers on Page 4 act as your 'calculator'. When you click Mumbai + Electric Bike + 2025, all four KPI cards instantly show that specific scenario. This is the Power BI equivalent of the slider in the HTML dashboard.

SECTION 5 — Theme & Formatting

Apply a Dark Theme

View → Themes → Browse for Themes → paste this JSON into a .json file and load it

```
{
  "name": "GigWorker Dark",
  "dataColors": ["#4ECB8D", "#E8C547", "#E05C5C", "#5B9BD5", "#9B72CF", "#E8834A"],
  "background": "#0D0F14",
  "foreground": "#E8EAF2",
  "tableAccent": "#E8C547"
}
```

Note: Save the JSON above as 'gig_dark_theme.json', then load it via **View → Themes → Browse for Themes**.

Key Format Settings Per Visual

Where to set	Value to use
Format → Page → Canvas background	#0D0F14 (100% transparency off)
Format pane → General → Background	#1A1E2A Border: #252A38
Format → Title → Font	Segoe UI Semibold, 11pt, white
Format → Data labels → Font	Segoe UI, 9pt
Format → Gridlines → Colour	#252A38 (very subtle)
Format → Callout Value → Font size	28–32pt, bold
Format → Legend → Zomato	#E05C5C (red)
Format → Legend → Zepto	#9B72CF (purple)
Format → Legend → Blinkit	#E8C547 (gold)
Format → Legend → Swiggy	#E8834A (orange)
Format → Legend → Dunzo	#4ECB8D (green)
Format → Legend → Mumbai	#5B9BD5 Delhi: #9B72CF Bangalore: #E8834A

SECTION 6 — Interview DAX Questions & Answers

These are the most likely DAX questions an interviewer will ask when they see this Power BI dashboard.

Q1: What is the difference between a Measure and a Calculated Column?

A Measure is calculated on the fly based on filter context — it changes when you select a slicer. A Calculated Column is computed row by row when the data loads and is stored in the table. In this project, `breakeven_days` is a Calculated Column because it uses row-level data. Avg Breakeven Days is a Measure because it aggregates across whatever the current filter shows.

Q2: What is filter context in DAX?

Filter context is the set of filters active at any point — from slicers, visuals, row/column headers. When I select Mumbai in the city slicer, the filter context restricts all measures to Mumbai rows only. My Real Wage Index measure uses `CALCULATE` to override filter context and always look at 2021 as the base year, regardless of what year the user has selected.

Q3: What does `CALCULATE` do?

`CALCULATE` evaluates a measure or expression in a modified filter context. In my Real Wage Index measure I use `CALCULATE(AVERAGE(real_net_inr), year=2021)` to force the 2021 base calculation even when the user has 2024 or 2025 selected. Without `CALCULATE` I could not fix the base year — the average would just reflect whatever year is currently filtered.

Q4: Explain `DIVIDE` vs `/`

`DIVIDE(numerator, denominator, alternate_result)` handles division by zero gracefully — it returns the alternate result or `BLANK` instead of an error. Using `/` directly will throw an error if the denominator is zero. I always use `DIVIDE` for percentage calculations like take-home rate to avoid breaking the visual when there is no data for a particular filter combination.

Q5: What is `SUMMARIZE`?

`SUMMARIZE` creates a summary table with grouping columns and aggregations — similar to `GROUP BY` in SQL. I used it in the Best Platform Net measure to first group by platform, calculate average net per platform, then take the maximum across all platforms. It is one of the most powerful DAX table functions for creating intermediate aggregations inside a measure.

Q6: What is the difference between `ALL` and `ALLEXCEPT`?

`ALL` removes all filters from a table or column. `ALLEXCEPT` removes all filters except the ones you specify. If I wanted a platform's share of total earnings I would use `DIVIDE(SUM(net), CALCULATE(SUM(net), ALL(Monthly_Earnings[platform])))` — the `ALL` removes the platform filter so the denominator is always the grand total regardless of which platform is selected.

Q7: How would you create a running total in DAX?

Using `CALCULATE` with `FILTER` and `ALL` combined with a date comparison. For example: Running Total = `CALCULATE([Total Net Earnings], FILTER(ALL(Date_Table), Date_Table[sort_order] <= MAX(Date_Table[sort_order])))`. This keeps accumulating as you move along the time axis.

Q8: What is a What-If Parameter?

A What-If Parameter creates a slicer-driven variable in Power BI. I could use it for the commission rate override in the break-even calculator — Modeling → New Parameter → name it Commission Override, range 17 to 27, step 0.5. Power BI creates a measure called Commission Override Value automatically, and I can reference it in my break-even DAX to recalculate dynamically as the user drags the slider. This replicates the commission slider I built in the HTML dashboard.

Q9: How did you handle the date table?

I created a separate Date_Table CSV with a year_month key that joins to Monthly_Earnings on year_month. I marked it as the Date Table in Power BI. This is best practice because it gives Power BI time intelligence context and ensures the sort_order column controls the axis sequence correctly. Without a proper date table, Power BI would alphabetically sort year_month strings instead of chronologically.

Q10: What is the difference between AVERAGE and AVERAGEX?

AVERAGE works on a column directly. AVERAGEX iterates over a table row by row, evaluates an expression for each row, then averages the results. For breakeven_days I could write `AVERAGEX(Monthly_Earnings, Monthly_Earnings[fixed_costs_inr] / Monthly_Earnings[net_per_day])` — this computes break-even for each row then averages. It is more flexible when the expression involves multiple columns from the same row.

SECTION 7 — Quick Reference Cheatsheet

Column Names You Will Use Most

Table	Column	Type	Notes
Monthly_Earnings	gross_earnings_inr	Whole Number	Main measure input
Monthly_Earnings	net_earnings_inr	Whole Number	Main measure input
Monthly_Earnings	platform_commission_pct	Decimal	Commission %
Monthly_Earnings	real_net_inr	Decimal	Pre-computed CPI-adjusted net
Monthly_Earnings	fixed_costs_inr	Decimal	Pre-computed: fuel+phone+maint
Monthly_Earnings	breakeven_days	Decimal	Pre-computed break-even
Monthly_Earnings	take_home_pct	Decimal	Pre-computed: net/gross x 100
Monthly_Earnings	is_monsoon	Text	Monsoon / Non-Monsoon
Monthly_Earnings	rain_bucket	Text	No Rain / Light / Heavy / Extreme
Monthly_Earnings	year_month	Text	Join key to Date_Table
Monthly_Earnings	year	Whole Number	Filter for year slicers
Worker_Profiles	vehicle_type	Text	Electric Bike / Petrol Bike etc
Worker_Profiles	engine_type	Text	Electric / ICE
Date_Table	sort_order	Whole Number	Use to sort months correctly
Date_Table	is_monsoon	Text	Monsoon flag for axis colour
City_Year_Summary	real_wage_index_2021_100	Decimal	Real wage index — Page 3

45-Minute Build Order

Time	Task
0–5 min	Import all 6 CSV files via Get Data → Text/CSV
5–10 min	Set data types in Power Query. Close & Apply.
10–15 min	Build all 5 relationships in Model View
15–25 min	Create all 20 DAX measures in a Measures table
25–30 min	Apply dark theme JSON
30–38 min	Build Pages 1 and 2 visuals
38–43 min	Build Pages 3 and 4 visuals
43–45 min	Add slicers, format titles, check all visual interactions

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